

$$1. \left(\frac{1}{2} \arcsin x + \frac{x}{2} \sqrt{1-x^2}\right)' =$$

$$2. (x^a)' =$$

$$3. (a^x)' =$$

$$4. (e^x)' =$$

$$5. (\log_a x)' =$$

$$6. (\ln |x|)' =$$

$$7. (\sin x)' =$$

$$8. (\cos x)' =$$

$$9. (\tan x)' =$$

$$10. (\cot x)' =$$

$$11. (\sec x)' =$$

$$12. (\csc x)' =$$

$$13. (\ln |\cos x|)' =$$

$$14. (\ln |\sin x|)' =$$

$$15. (\ln |\sec x + \tan x|)' =$$

$$16. (\ln |\csc x + \cot x|)' =$$

$$17. (\arcsin x)' =$$

$$18. (\arccos x)' =$$

$$19. (\arctan x)' =$$

$$20. (\operatorname{arccot} x)' =$$

$$21. [\ln(x + \sqrt{x^2 + 1})]' =$$

$$22. [\ln(x + \sqrt{x^2 - 1})]' =$$

$$23. \int x^k \, dx =$$

$$24. \int \frac{1}{x} \, dx =$$

$$25. \int e^x \, dx =$$

$$26. \int a^x \, dx =$$

$$27. \int \sin x \, dx =$$

$$28. \int \cos x \, dx =$$

$$29. \int \tan x \, dx =$$

$$30. \int \cot x \, dx =$$

$$31. \int \frac{1}{\cos x} \, dx =$$

$$32. \int \frac{1}{\sin x} \, dx =$$

$$33. \int \frac{1}{\cos^2 x} \, dx =$$

$$34. \int \frac{1}{\sin^2 x} \, dx =$$

$$35. \int \sec x \tan x \, dx =$$

$$36. \int \csc x \cot x \, dx =$$

$$37. \int \frac{1}{1+x^2} \, dx =$$

$$38. \int \frac{1}{a^2+x^2} \, dx =$$

$$39. \int \frac{1}{\sqrt{1-x^2}} \, dx =$$

$$40. \int \frac{1}{\sqrt{a^2-x^2}} \, dx =$$

$$41. \int \sqrt{1-x^2} \, dx =$$

$$42. \int \sqrt{a^2-x^2} \, dx =$$

$$43. \int \frac{1}{\sqrt{x^2+1}} \, dx =$$

$$44. \int \frac{1}{\sqrt{x^2+a^2}} \, dx =$$

$$45. \int \frac{1}{\sqrt{x^2-1}} \, dx =$$

$$46. \int \frac{1}{\sqrt{x^2-a^2}} \, dx =$$

$$47. \int \sqrt{1+x^2} \, dx =$$

$$48. \int \frac{1}{x^2-a^2} \, dx =$$

$$49. \int \frac{1}{a^2-x^2} \, dx =$$

$$50. \int \sin^2 x \, dx =$$

$$51. \int \cos^2 x \, dx =$$

$$52. \int \tan^2 x \, dx =$$

$$53. \int \cot^2 x \, dx =$$

$$54. \int \tan^n x \, dx =$$